

EXPERIMENT – 5

PERFORM VARIOUS CONFIGURATIONS OF IPTABLES FIREWALL IN LINUX

AIM

To study and perform various configurations of the IPTABLES firewall in Linux to control incoming and outgoing network traffic.

REQUIREMENTS

Linux Operating System (Ubuntu / Kali Linux)
Root or sudo privileges
IPTABLES firewall utility

PROCEDURE

STEP 1: OPEN TERMINAL

Boot Linux system
Open Terminal
Switch to root user or use sudo
sudo su

STEP 2: VIEW CURRENT IPTABLES RULES

sudo iptables -L

STEP 3: FLUSH EXISTING RULES

sudo iptables -F

STEP 4: SET DEFAULT POLICIES

sudo iptables -P INPUT DROP
sudo iptables -P FORWARD DROP
sudo iptables -P OUTPUT ACCEPT

STEP 5: ALLOW LOCALHOST TRAFFIC

sudo iptables -A INPUT -i lo -j ACCEPT

STEP 6: ALLOW ESTABLISHED CONNECTIONS

sudo iptables -A INPUT -m state --state ESTABLISHED,RELATED -j ACCEPT

STEP 7: ALLOW SSH TRAFFIC (PORT 22)

sudo iptables -A INPUT -p tcp --dport 22 -j ACCEPT

STEP 8: ALLOW HTTP AND HTTPS TRAFFIC

sudo iptables -A INPUT -p tcp --dport 80 -j ACCEPT
sudo iptables -A INPUT -p tcp --dport 443 -j ACCEPT

STEP 9: BLOCK ICMP (PING REQUESTS)

sudo iptables -A INPUT -p icmp -j DROP

STEP 10: VERIFY IPTABLES RULES

sudo iptables -L

STEP 11: SAVE FIREWALL RULES

For Kali / Debian / Ubuntu (with iptables-persistent):
sudo iptables-save > /etc/iptables/rules.v4

OUTPUT

IPTABLES rules were successfully applied
Incoming and outgoing traffic was controlled
Rules verified using iptables -L

```
(kali@kali)-[~]
$ sudo iptables -L
Chain INPUT (policy DROP)
target    prot opt source                destination
ACCEPT    all  --  anywhere              anywhere
ACCEPT    all  --  anywhere              anywhere
ACCEPT    tcp  --  anywhere              anywhere    state RELATED,ESTABLISHED
ACCEPT    tcp  --  anywhere              anywhere    tcp dpt:ssh
ACCEPT    tcp  --  anywhere              anywhere    tcp dpt:http
ACCEPT    tcp  --  anywhere              anywhere    tcp dpt:https
DROP      icmp --  anywhere              anywhere

Chain FORWARD (policy DROP)
target    prot opt source                destination

Chain OUTPUT (policy ACCEPT)
target    prot opt source                destination
```

RESULT

Thus, various configurations of the IPTABLES firewall were successfully performed in Linux to control network traffic and enhance system security.